

Three-uniform word representations of all bipartite graphs with parts of size six

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Abstract

Every finite simple bipartite graph with two specified parts of size six has a word representation in which each vertex occurs exactly three times. We give a computer-assisted proof covering all 251,610 orbits under independent relabeling of the two parts. Component and false-twin reductions, together with four stated results of Mozhui and Krishna, cover 242,933 orbits. The remaining 8,677 orbits have explicit length-36 word certificates, checked on all 66 vertex pairs. Exact canonicalization and Burnside’s lemma certify completeness of the orbit list without relying on the graph generator. The conclusion concerns the full $6 + 6$ layer, including isolated vertices and repeated neighborhoods. It makes no assertion about larger equal-even parts; priority of this finite resolution remains unverified.

1 Statement and context

For a word w on a vertex set V , let $w|_{\{x,y\}}$ be the word obtained by deleting letters other than x and y . Two distinct letters *alternate* when no two consecutive letters of this projection are equal. A word is *k-uniform* if every vertex occurs exactly k times. It represents a finite simple graph $G = (V, E)$ when, for every distinct $x, y \in V$,

$$xy \in E \iff w|_{\{x,y\}} \text{ alternates.}$$

The representation number $\mathcal{R}(G)$ is the least such positive integer k .

Theorem 1. *Let $G = (A \sqcup B, E)$ be any finite simple bipartite graph with $|A| = |B| = 6$ and $E \subseteq A \times B$. There exists a word of length 36 in which each vertex occurs exactly three times and two vertices alternate if and only if they are adjacent. In particular, $\mathcal{R}(G) \leq 3$.*

There are no connectivity, degree, or distinct-neighborhood assumptions in Theorem 1. The equivalence includes the 30 within-part pairs as well as the 36 cross-part pairs.

Mozhui and Krishna [1] study the bound $\mathcal{R}(G) \leq \lceil (|A| + |B|)/4 \rceil$, attributed there to Glen and coauthors for $|A| + |B| \geq 9$. Their general results leave the equal-even family and establish sufficient neighborhood-inclusion conditions within it. For parts of size six, their general bound is four and their two inclusion criteria give three. The contribution here is a complete finite verification after those criteria, with explicit positive certificates for every residual orbit. This is more than testing a sample of graphs or checking only graphs that satisfy those criteria.

As adjacent methodology, Zeng’s finite verification of a line-graph signature inequality [2] also combines exhaustive isomorphism classes with exact checks. That work is cited for computational graph verification, not as a result on word representations. Our comparison with the word-representation literature is bounded by the sources inspected; the limitation concerning a relevant 2026 paper is stated in Section 5.

2 Theorem dependencies and elementary reductions

Write $N(v)$ for the open neighborhood of v . A graph is *reduced* if distinct vertices have distinct neighborhoods. We use the following four results of Mozhui and Krishna, with the connected and reduced convention on printed page 2 of [1]. In the first two items, $m \leq n$ are the part sizes.

- (D1) Corollary 1, page 6: $\mathcal{R}(G) \leq 1 + \lceil m/2 \rceil$.
- (D2) Theorem 3, page 7: if $5 \leq m \leq n$ and m is odd, then $\mathcal{R}(G) \leq \lceil m/2 \rceil$.
- (D3) Theorem 4, page 9, specialized to a connected reduced $6 + 6$ graph: if the neighborhood-inclusion poset in either part has a chain of three vertices, then $\mathcal{R}(G) \leq 3$.
- (D4) Theorem 5, page 13, in the same $6 + 6$ setting: if that poset has height two and two vertex-disjoint chains of size two, then $\mathcal{R}(G) \leq 3$.

Here the strict order within a part is $x \prec y$ if $N(x) \subsetneq N(y)$, and height is the maximum number of vertices in a chain. The final two items are the $k = 3$ cases of results for a part of size $2k$, $k \geq 3$. These four statements are external theorem dependencies, whose general proofs are not reproduced here.

Lemma 2. *The property of admitting a 3-uniform representing word has the following closure properties.*

- (i) *A representing word of uniformity at most three can be increased to uniformity three.*
- (ii) *Disjoint unions of graphs with this property have the property.*
- (iii) *Adding a false twin preserves the property, where a false twin of x is a new nonadjacent vertex y with $N(y) = N(x)$.*

Proof. Let w be k -uniform, and let p list its distinct letters in order of first occurrence. Prepending p raises uniformity to $k + 1$. On an edge, the old projection is $(xy)^k$ or $(yx)^k$, and the prefix adds xy or yx , respectively. On a nonedge, the old nonalternating projection survives as a suffix. Iteration proves (i).

For (ii), concatenate 3-uniform words on the disjoint component alphabets. An intracomponent projection is unchanged. A pair from different components projects to x^3y^3 or y^3x^3 , which does not alternate. A singleton component has word x^3 .

For (iii), replace the first, second, and third occurrences of x by xy , xy , and yx , respectively. Old projections are unchanged. For $z \neq x, y$, the projection to y, z is the old projection to x, z with x renamed y . The projection to x, y is $xyxyyx$, which does not alternate. Thus the new vertex has exactly the required adjacencies. $\square$

Proposition 3. *Every disconnected $6 + 6$ bipartite graph, and every connected $6 + 6$ bipartite graph with a repeated neighborhood within a part, has a 3-uniform representing word.*

Proof. Consider first a nontrivial connected bipartite graph whose smaller part has size at most five. Successively delete one of any pair of vertices with equal neighborhoods within a part. Deleting a false twin preserves connectedness: a walk through the deleted vertex can use the surviving twin and then be shortened to a path. The process terminates at a connected reduced graph, still with smaller part size $m \leq 5$. Indeed, neighborhoods of vertices in opposite parts are nonempty subsets of disjoint sets, so cannot be equal. For $m \leq 4$ apply (D1), and for $m = 5$ apply (D2). Lemma 2 pads the resulting word to uniformity three and restores deleted twins in reverse order.

Each nontrivial component of a disconnected $6 + 6$ graph has smaller part at most five: a component using all six vertices from both parts would be the entire graph. Treat these components as above, handle singletons explicitly, and concatenate their words. Finally, a connected $6 + 6$ graph with a repeated neighborhood loses at least one vertex from one part under twin deletion, so its smaller part becomes at most five. Apply the same argument and restore the twins. $\square$

These constructions concern the threshold three directly; they do not require identities for representation number at uniformity one.

3 Completeness of the finite graph list

Label $A = \{0, \dots, 5\}$ and $B = \{6, \dots, 11\}$. A graph is a binary 6×6 cross-adjacency matrix. Independent relabeling acts by $S_6 \times S_6$. Throughout the enumeration, parts are specified: neither transposition nor bipartite complementation is used to identify or discard representatives.

Lemma 4. *There are exactly 251,610 orbits of binary 6×6 matrices under $S_6 \times S_6$.*

Proof. For a partition λ of six, let c_i be the multiplicity of the part i and set $z_\lambda = \prod_i i^{c_i} c_i!$. There are $6!/z_\lambda$ permutations of cycle type λ . A row cycle of length a and a column cycle of length b have $\gcd(a, b)$ orbits on their ab matrix positions: each orbit has length $\text{lcm}(a, b)$. A fixed binary matrix is constant on each position orbit. Burnside's lemma therefore gives

$$\#\text{orbits} = \sum_{\lambda, \mu \vdash 6} \frac{2^{\sum_{a \in \lambda, b \in \mu} \gcd(a, b)}}{z_\lambda z_\mu} = \frac{130,434,624,000}{(6!)^2} = 251,610. \quad (1)$$

The sums over parts count multiplicities. The numerical evaluation uses exact integers or fractions; it was also recomputed by enumerating all 720 permutations to obtain the cycle-type multiplicities. $\square$

For completeness, counting the supplied records is not enough: their orbits must be distinct. Define a canonical key for a binary matrix as follows. Enumerate all 720 column permutations. For each, view the rows as six-bit integers, sort them, and pack the sorted list into a 36-bit integer. Take the minimum over the column permutations. Every candidate is in the original orbit. Sorting removes row order, while enumeration exhausts column order. Hence two matrices have the same key if and only if they lie in the same fixed-part orbit.

The supplied list contains 251,610 valid matrices and their computed keys are pairwise distinct. Lemma 4 then proves that the list covers every orbit exactly once. This argument does not require trusting the original graph generator. A second implementation permuted rows and sorted columns, independently confirming distinctness for the entire list. The second algorithm serves as an additional check; its packed keys need not equal those of the first algorithm.

4 Classification and positive word certificates

Process the complete list in the following order. First test connectedness. Among connected graphs, test for equal neighborhoods within each part. For the remaining reduced graphs, form the strict neighborhood-inclusion order in each part. A chain of size three invokes (D3). If no such chain exists in a part but two disjoint inclusion pairs do, its height is exactly two and (D4) applies. A graph passing either criterion in either part is covered. All other graphs

Disjoint category, in testing order	Orbits
Disconnected	57,407
Connected with a within-part false-twin pair	110,152
Connected, reduced, covered by (D3) or (D4)	75,374
Residual connected reduced graphs	8,677
Total	251,610

Table 1: Exact classification of the complete fixed-part orbit list.

form the residual list. Failure of these sufficient conditions alone gives no lower bound on $\mathcal{R}(G)$.

Both a set-based implementation and a separate bitmask implementation reproduced Table 1. In the latter, the inclusion height is computed by dynamic programming on ordered neighborhood masks, and disjoint inclusion endpoints are checked at height two. Both reproduced the residual list entry for entry, including its order.

For every residual graph a positive certificate gives a word of length 36. The certificate file binds each word to a unique zero-based index and the exact graph encoding at that index in the residual list. The checker verifies the index range, absence of duplicate indices, equality of graph encodings, word length, alphabet $\{0, \dots, 11\}$, and three occurrences of each vertex. For each of the $\binom{12}{2} = 66$ pairs $x < y$, it then checks

$$(w|_{\{x,y\}} \in \{(xy)^3, (yx)^3\}) \iff xy \in E. \quad (2)$$

Finally, the set of verified indices must be the entire residual index set. All 8,677 words pass, giving 572,682 pair checks. There are no missing or unresolved records. In particular, checking only cross-part pairs would be insufficient, and is not the check used here.

The graph format is directly decodable. Each record starts with the graph6 size character for twelve vertices. The next eleven characters, after subtracting 63 from their character codes and expanding each to six bits, encode the 66 potential edges in the order

$$(0, 1), (0, 2), (1, 2), (0, 3), (1, 3), (2, 3), \dots, (10, 11).$$

The decoder also checks the alphabet and that no encoded edge is within a part. A separate solver-free verifier tested alternation by scanning for consecutive equal projected letters, obtaining the same full result. Deliberately invalid words, including words making all pairs alternate and words making none alternate, were rejected as diagnostic controls.

Proof of Theorem 1. An arbitrary graph in the stated domain lies in an orbit of the complete list certified in Section 3. Its representative belongs to exactly one row of Table 1. Proposition 3 handles the first two rows. Dependencies (D3) and (D4), together with padding if necessary, handle the third. For the last row, the corresponding checked word satisfies (2) on all vertex pairs. Transport the resulting word along the part-preserving isomorphism to the original graph. Relabeling preserves both adjacency and pairwise alternation, proving the assertion. $\square$

5 Reproduction and limitations

The accompanying supplement contains the full orbit list, canonical keys, residual list, all positive words, and verification sources. Its `README.md` identifies these data files and gives commands that require a C++17 compiler and Python’s standard library. The checks replay full coverage rather than a sampled prefix. Exact canonicalization, the orbit-count identity, classification, and direct word verification are separate obligations.

Word discovery used Z3 with integer occurrence positions $p(v, 0) < p(v, 1) < p(v, 2)$, all 36 positions distinct and in $\{0, \dots, 35\}$. For every vertex pair, the disjunction of the two strict alternating interleavings was constrained to agree with adjacency. The search produced positive words only; its completeness and any negative solver response are unnecessary for the proof once every required word has been checked directly. The verification relies on ordinary exact C++ and Python execution and the four stated external theorems. No proof-assistant formalization is claimed.

The finite theorem does not establish the bound for larger equal-even part sizes. The closest inspected primary text is the June 2025 version of Mozhui and Krishna [1]. Its discussion of the remaining general family does not establish the present open status of the $6 + 6$ layer. The full text of Hefty, Horn, Muir, and Owens [3] was not inspected; only its bibliographic record was available in the literature dossier. No assertion is made about whether that paper contains an equivalent finite conclusion. As of 9 September 2026, priority and absence of a later equivalent result remain unverified.

Computational and AI assistance. AI assistance was used to develop the argument, generate computational code, search for representing words, prepare the manuscript, and conduct a separate verification pass using independently implemented checkers. That pass recomputed all orbit distinctions, the classification, and all word checks. The evidence for the theorem consists of the stated mathematical dependencies, the reductions, and the exact finite verification; agreement between AI-assisted passes is not itself a proof.

References

- [1] Khyodeno Mozhui and K. V. Krishna, *On the Conjecture of the Representation Number of Bipartite Graphs*, arXiv:2506.01057v1 (2025). <https://arxiv.org/abs/2506.01057v1>.
- [2] Zijian Zeng, *Exact Verification of the Line-Graph Signature Inequality at Order Nine*, SSRN working paper (2026). <https://doi.org/10.2139/ssrn.7386040>.
- [3] Zion Hefty, Paul Horn, Colby Muir, and Andrew Owens, Word-representation numbers of graphs: Bottlenecks and bounds, *Journal of Combinatorial Theory, Series A* **223** (2026), 106215. <https://doi.org/10.1016/j.jcta.2026.106215>.